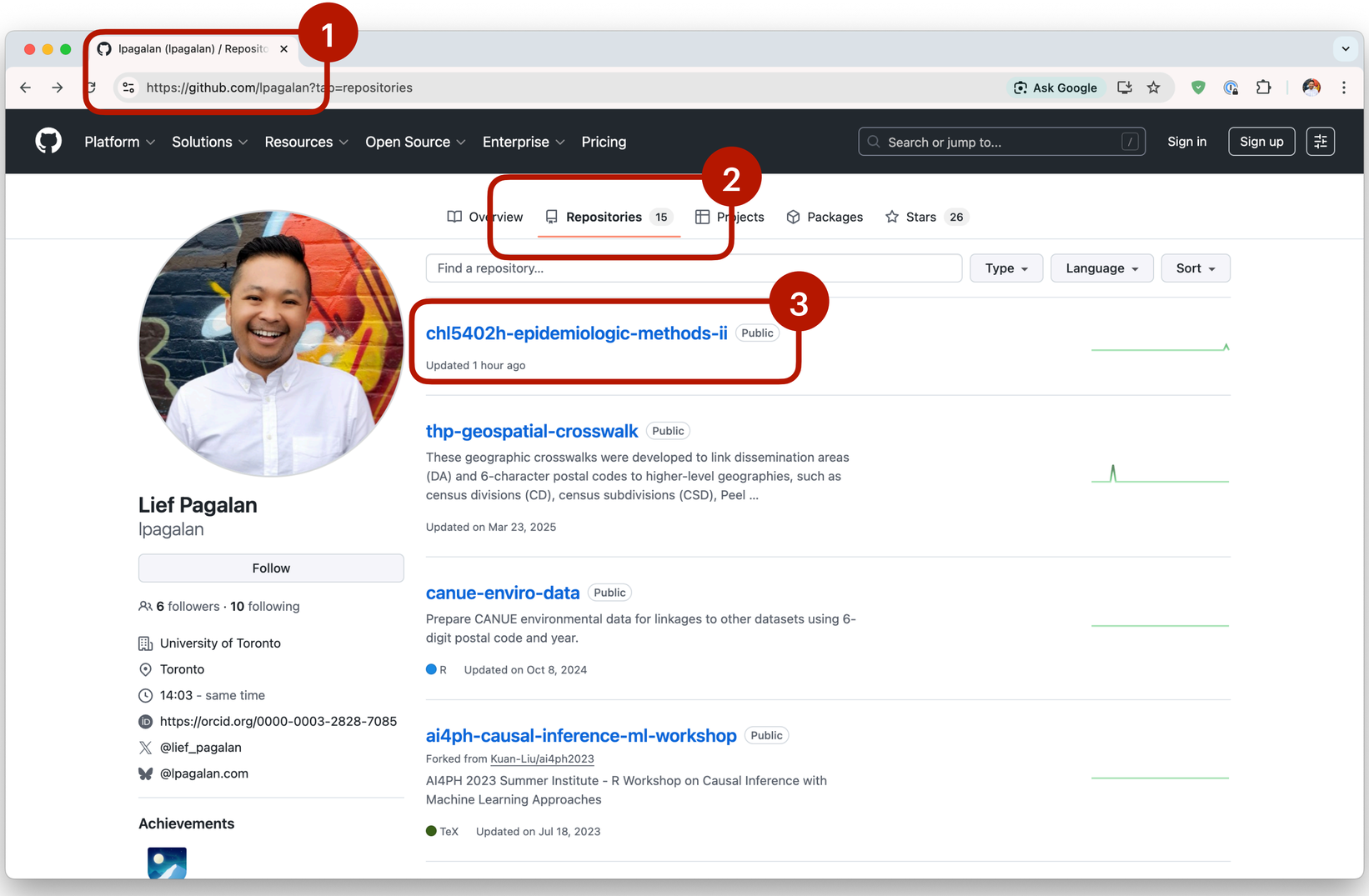
Confounding and Effect Modification

Tutorial 04

CHL5402H — March 3, 2026

Online Materials



GitHub

https://github.com/lpagalan

Agenda

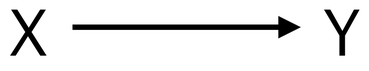
- Attendance
- Confounding and Effect Modification Summary
- Activity 1: Interpreting Regression Tables
- Activity 2: Spousal Loss Study—Confounders and Effect Modifiers

Learning Goals

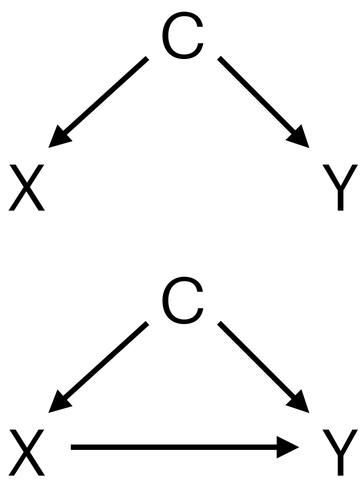
- Differentiate between confounding, mediation, and effect modification
- Interpret regression results and coefficients
- Assess for confounding and effect mediation

Confounding and Effect Modification Summary

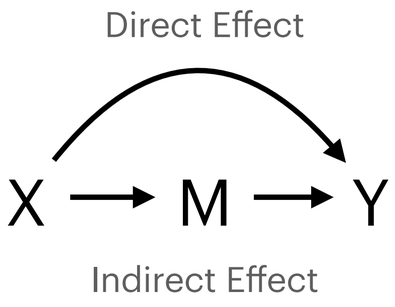
Confounding, Mediation, Effect Modification



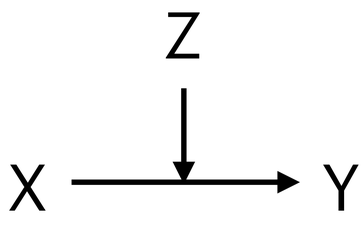
Focal Association



Confounding

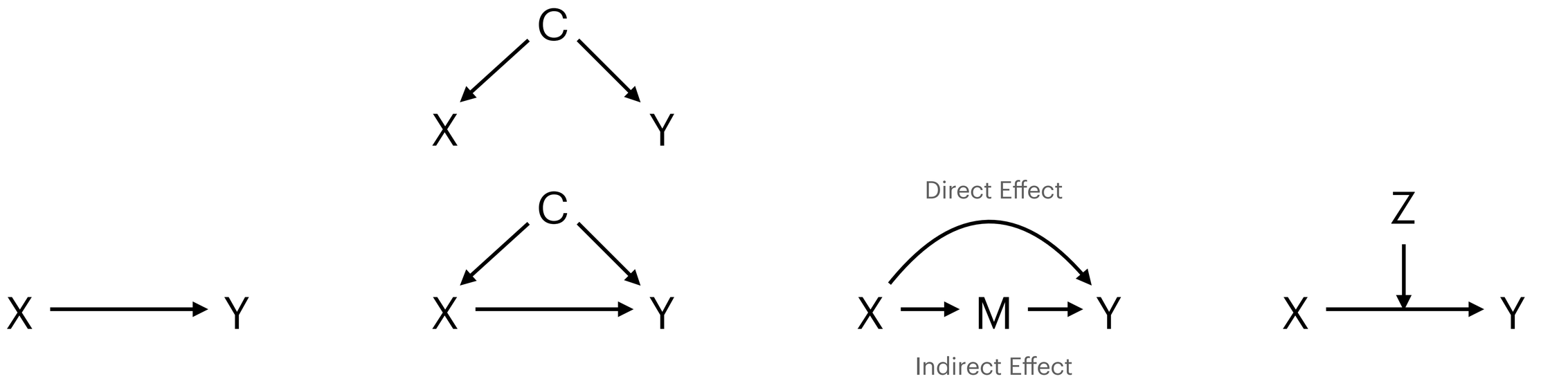


Mediation



Effect Modification

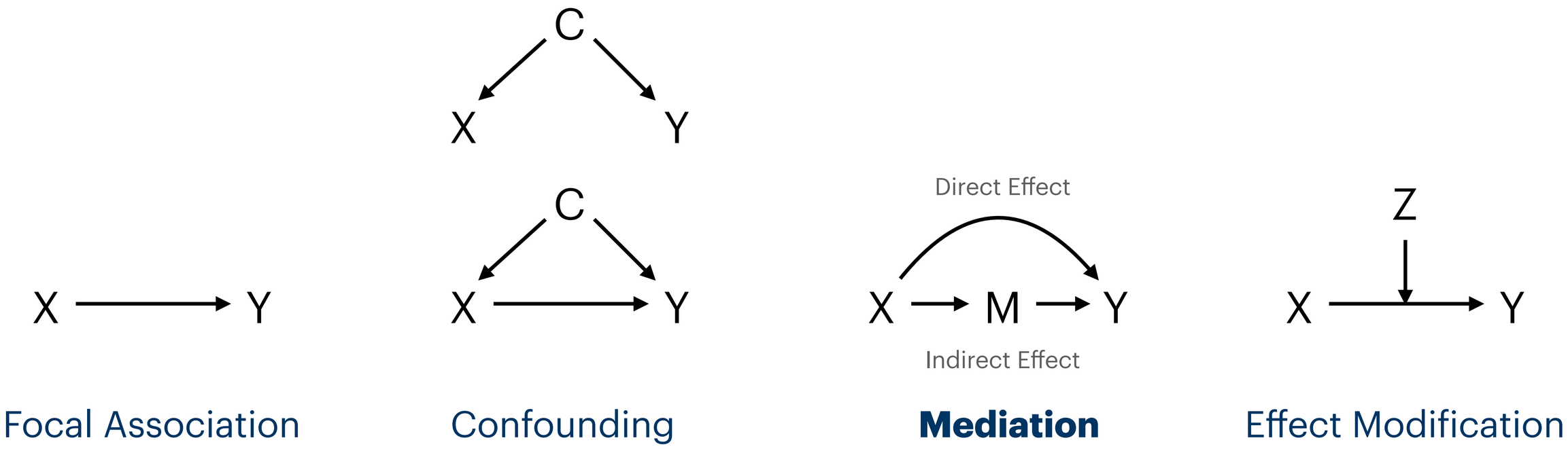
Confounding, Mediation, Effect Modification



Focal Association | **Confounding** | Mediation | Effect Modification

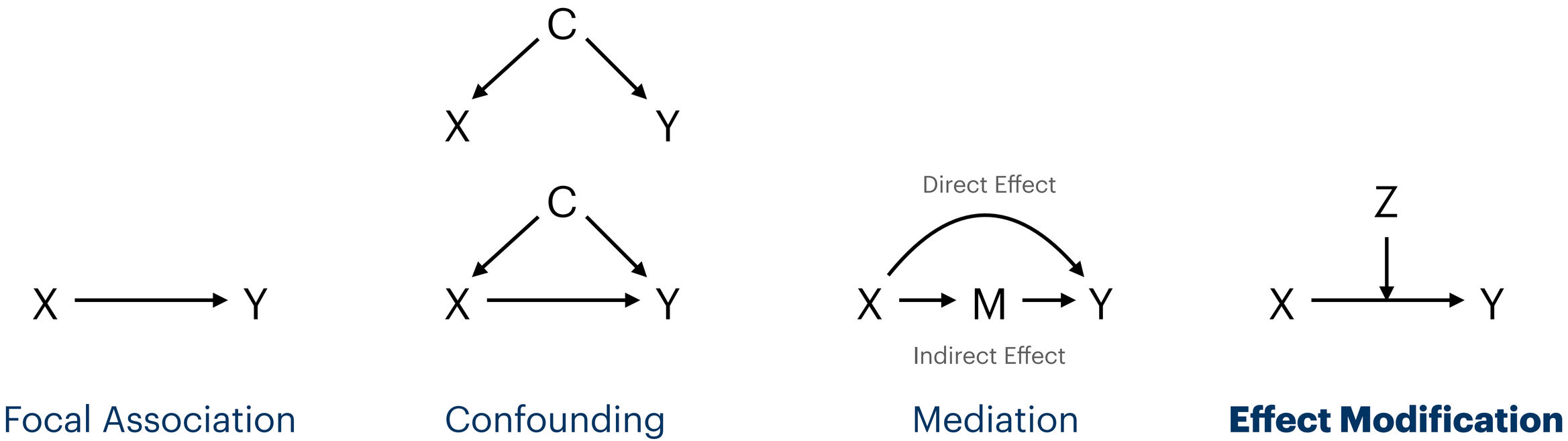
- A non-causal association between a given exposure and an outcome
- Observed association as a result of the influence of a third variable
- Confounding variable must be related to both the risk factor and outcome

Confounding, Mediation, Effect Modification



- Decompose pathways that could explain the effect of an exposure on an outcome
- Direct effect of X→Y and indirect effect of X on Y through mediator M
- Total effect of exposure on outcome: direct + indirect effects

Confounding, Mediation, Effect Modification



- Effect modification occurs when the effect of the exposure on the outcome differs depending on whether another variable (the effect modifier) is present

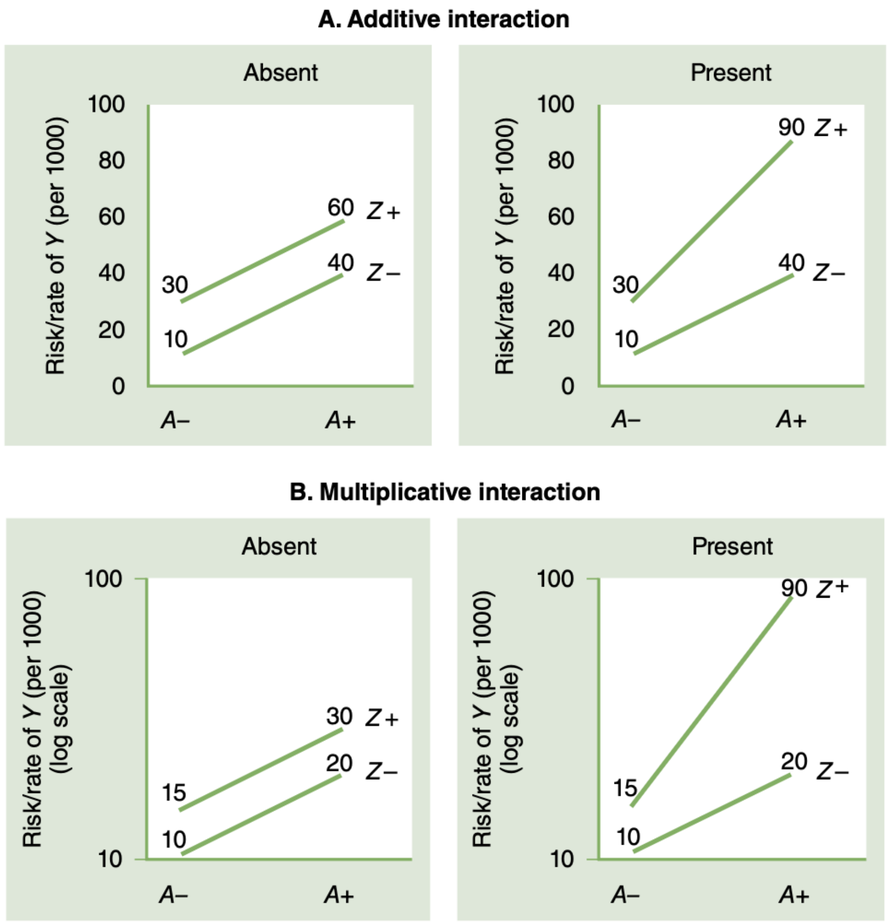


FIGURE 6-2 Assessment of interaction by means of graphs. For additive interaction (A), an arithmetic scale should be used on the ordinate (slopes represent absolute differences). Interaction is absent on the left panel because the absolute difference between *A*+ and *A*− is the same regardless of the presence of *Z* (60 − 30 = 40 − 10 = 30 per 1000). Interaction is present on the right panel because the absolute difference between *A*+ and *A*− is greater when *Z* is present (90 − 30 = 60 per 1000) than when *Z* is absent (40 − 10 = 30 per 1000). For multiplicative interaction (B), a logarithmic scale should be used on the ordinate (slopes represent relative differences). Interaction is absent on the left panel because the relative difference between *A*+ and *A*− is the same regardless of the presence of *Z* (30/15 = 20/10 = 2). Interaction is present on the right panel because the relative difference between *A*+ and *A*− is higher when *Z* is present (90/15 = 6) than when *Z* is absent (20/10 = 2).

- Additive interaction (absolute difference)
- Multiplicative interaction (relative difference)

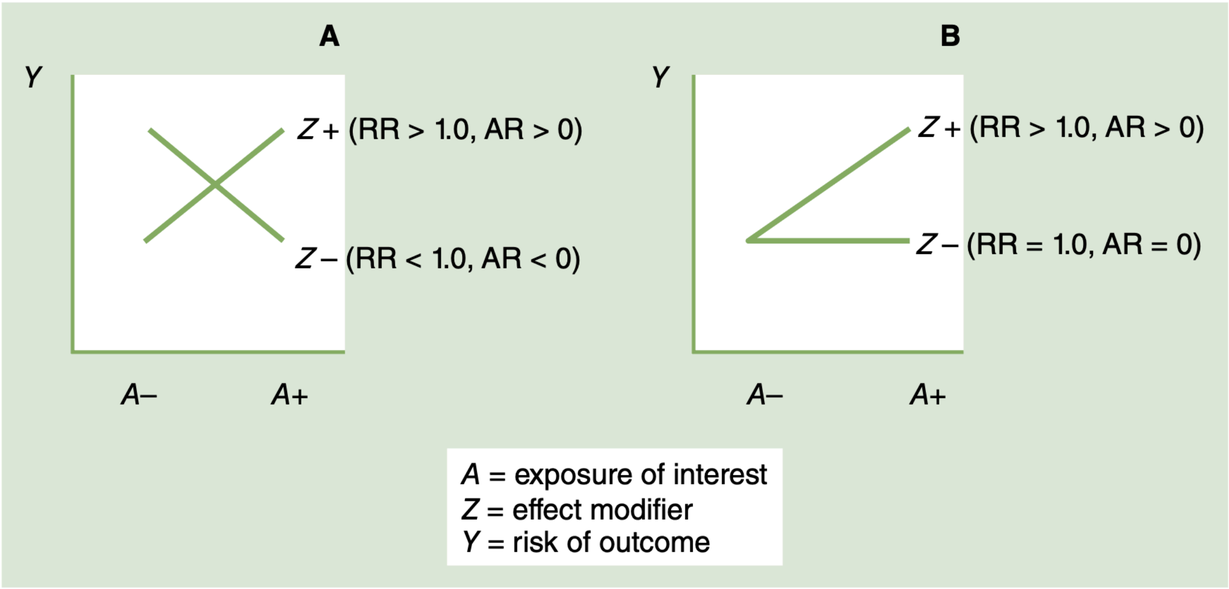


FIGURE 6-5 Qualitative interaction is always seen in both scales (additive and multiplicative) because when the relative risk (RR) > 1.0, the attributable risk (AR) > 0; when RR < 1.0, the AR < 0; and when RR = 1.0, the AR = 0. In (A), there is crossover; that is, $RR_{A+/A-} > 1.0$ and $AR_{A+/A-} > 0$ when *Z* (effect modifier) is present, and $RR_{A+/A-} < 1.0$ and $AR_{A+/A-} < 0$ when *Z* is absent. In (B), $RR_{A+/A-} > 1.0$ and $AR_{A+/A-} > 0$ when *Z* is present, and $RR_{A+/A-} = 1.0$ and $AR_{A+/A-} = 0$ when *Z* is absent.

- **Qualitative interaction:** When either the effects of A on the outcome Y are in opposite directions (crossover) according to the presence of the third variable Z or there is an association in one of the strata formed by Z but not in the other.
- **Quantitative interaction:** When the association between factor A and outcome Y exists and is of the same direction in each stratum formed by Z but the strength of the association varies across strata

Activity 1: Interpreting Regression Tables

TABLE 7-15 Multiple-regression models and interpretation of the regression coefficients $= b_1$.

	Model	Interpretation of b_1
Linear	$y = b_0 + b_1x_1 + b_2x_2 + \cdots + b_kx_k$	Increase in outcome y mean value (continuous variable) per unit increase in x_1, adjusted for all other variables in the model
Logistic	$\log(\text{odds}) = b_0 + b_1x_1 + b_2x_2 + \cdots + b_kx_k$	Increase in the log odds of the outcome per unit increase in x_1, adjusted for all other variables in the model
Cox	$\log(\text{hazard}) = b_0 + b_1x_1 + b_2x_2 + \cdots + b_kx_k$	Increase in the log hazard of the outcome per unit increase in x_1, adjusted for all other variables in the model
Poisson	$\log(\text{rate}) = b_0 + b_1x_1 + b_2x_2 + \cdots + b_kx_k$	Increase in the log rate of the outcome per unit increase in x_1, adjusted for all other variables in the model

e^{β} = Odds Ratio

e^{β} = Hazard Ratio

e^{β} = Rate Ratio

Interpreting Regression Tables

Table 2 Fallacy

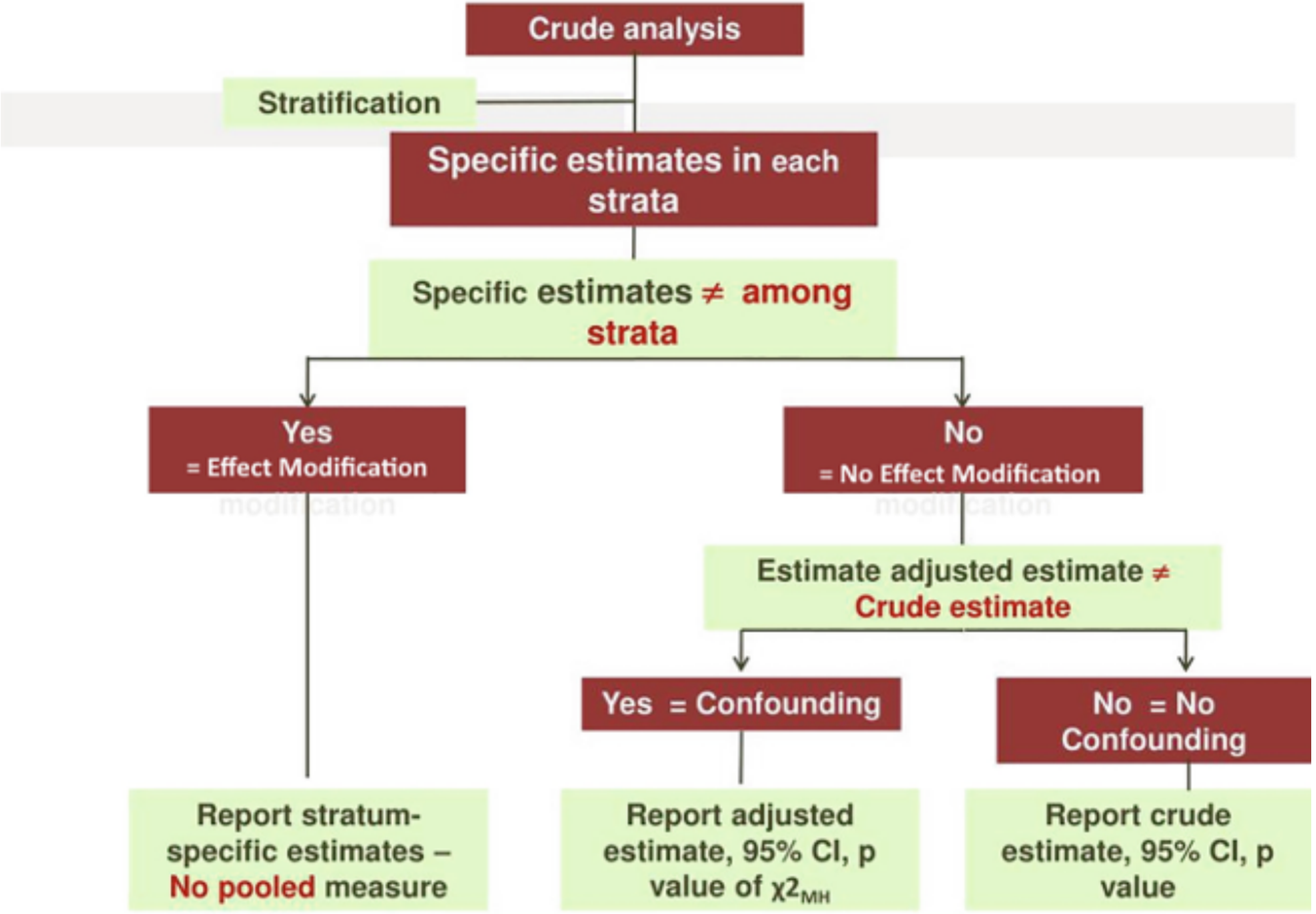
- **Descriptive vs. Causal:** Distinguish between mutually adjusted descriptive tables vs. a causal estimate adjusted for confounding
- **Causal Estimate:** A model should generally be built to answer a specific question about one primary exposure.
  - Coefficients from a multivariable regression model in a single table (typically "Table 2") should not be interpret as if they are all "adjusted" effects of equal standing
  - Interpretation of a coefficient depends entirely on the causal structure of the model—specifically whether a variable is an exposure, a confounder, or a mediator
  - A single model cannot simultaneously provide valid total effects for multiple variables if they have causal relationships with each other

Daniel Westreich, Sander Greenland, The Table 2 Fallacy: Presenting and Interpreting Confounder and Modifier Coefficients, American Journal of Epidemiology, Volume 177, Issue 4, 15 February 2013, Pages 292–298, https://doi.org/10.1093/aje/kws412

Interpreting Regression Tables

- Identify the focal exposure and additional covariates
- Identify where the predictor variable X is continuous, binary, categorical, etc.
- Interpret the measure of association in full sentences
  - Are you comfortable regardless of the form of regression (linear, logistic, survival, etc.) and regardless of the type of X variable?
- Which estimate(s) are most similar to the measure of association for your focal relationship of interest in your study?
- If none of these examples are exactly like your project association, what would you present?

Activity 2: Confounders and Effect Modifiers



Noncollapsibility

- Conditioning on non-confounding risk factors can cause the appearance of substantial confounding based on the change-in-effect approach.
- A statistical phenomenon where the adjusted effect does not equal the unadjusted effect, even in the absence of confounding.
- Mathematical property of certain link functions—most notably the logit (logistic regression and odds ratios).

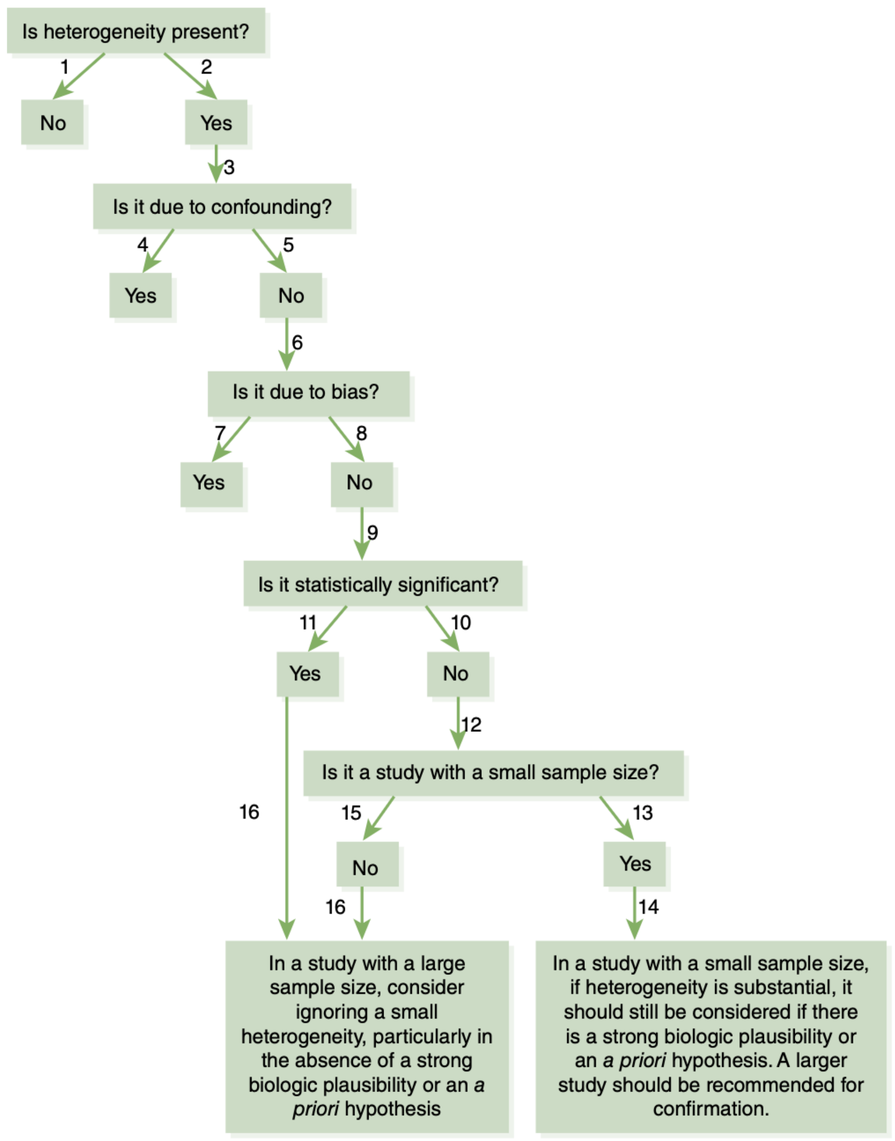


FIGURE 6-10 Evaluation of interaction in an epidemiologic study. If heterogeneity is present and not due to confounding or bias when it is statistically nonsignificant but strong, it should still be considered in the presence of strong biologic plausibility or an *a priori* hypothesis. A small but statistically significant heterogeneity detected in a study with a large sample size (e.g., a multicenter study) perhaps should be ignored in the absence of a strong biologic plausibility or an *a priori* hypothesis.

Interpreting Regression Tables

- What are the focal exposure and outcomes?
- Which other measures are being treated as confounders?
- Which other measures are being treated as effect modifiers?
- What is or are the final best measures of association that you would report that best characterize the estimate(s) of the causal association? Be prepared to defend your answer.